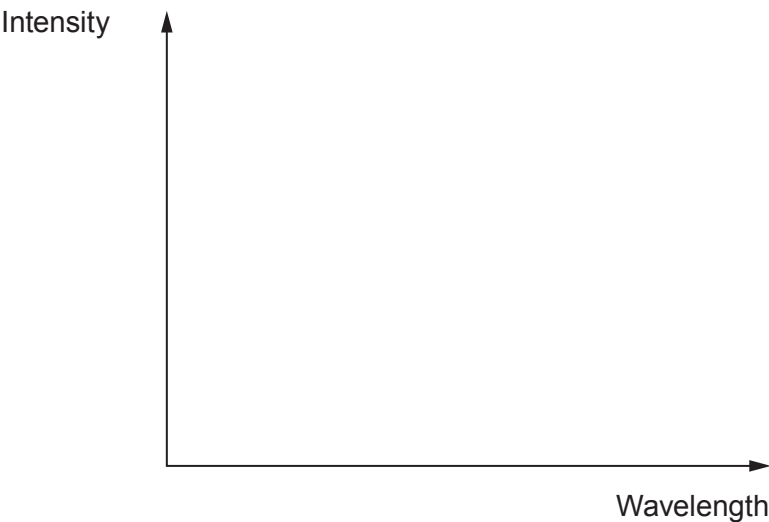


Option B – Medical Physics

7. An X-ray machine has a working potential difference of 75 000 V.
- (a) (i) Sketch a graph of intensity against wavelength for the resulting X-ray spectrum. Label the main features of this spectrum, including a value for the minimum wavelength. [2]

Space for calculation



- (ii) At the working potential difference the current in the tube is 120 mA and the efficiency of the X-ray machine is 0.7 %. Calculate the rate of production of heat. [2]

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Examiner
only

- (iii) A metal plate of thickness 1.4 mm is used to reduce the intensity of the X-rays produced to 60 % of the incident intensity. If a second identical plate is now also placed in the beam, calculate the new transmitted **percentage** intensity. [3]

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- (b) (i) Describe how the Doppler shift principle can be used to measure the speed of blood through an artery. [2]

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- (ii) Ultrasound of frequency 2 MHz was used to calculate the speed of blood and a Doppler shift of 200 Hz was detected. The measurement was taken at an angle of 40° to the direction of flow and the speed of ultrasound through the blood is 1500 m s⁻¹. Calculate the speed of blood flow. [2]

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Examiner
only

- (c) (i) Describe the properties of technetium–99m (Tc–99m) that make it such a good radioisotope in the effective diagnosis of medical problems. Justify your choice of properties. [3]

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- (ii) Explain clearly how a gamma camera is used to detect the gamma rays given off by a technetium–99m source. [3]

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- (iii) In positron emission tomography a positron annihilates an electron producing two photons of energy 0.511 MeV. By setting out your reasoning clearly, determine whether or not the value of the photon energy is correct. [3]

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